WASTEWISE-ML · FEASIBILITY ASSESSMENT

WasteWise-ml feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First feasibility assessment (kit 1.8 upgrade), closed at stage P3 with the open items named

WasteWise-ml feasibility assessment

Verdict: feasible on paper, with the data as the one Low-rated aspect. The hardware, the model's size, speed and energy, and the build are all well inside known limits and buildable with hand tools (WML-STM-001, WML-CAL-001, WML-BLD-001). Whether the model reaches its accuracy and coverage targets on real waste depends on about 10,000 consented field photos and a 2,000-image evaluation set that do not yet exist; building them is TRL 4 work with a partner and is on hold. The assessment is closed at stage P3 with that item open. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	One item on a mat under a phone held 440 mm above it on a clamped stand; the picture (586 x 439 mm) falls on the 700 x 450 mm mat with the stand outside it; every rig joint face to face and bolted; 115 constructability checks pass	High	The picture has 5.3 mm to spare front and back, 3.3 mm of it taken by the phone's placement in the tray after fix F5, so only lenses of 25.76 mm equivalent or longer fit (PR-019 at risk); a 420 mm lens height is proposed to Amish (D2); measured on the bought phone before the arm is cut	WML-STM-001 S1, S2, S6, S7; WML-CAL-001 sections 6 and 11; `python cad/src/model.py --check`
Materials	Plywood, 6063 aluminium tube, angle and sheet, painted and sealed hardboard, stainless fixings; all common stock	High	Hardboard swells if any face or edge is unsealed (fixed by F2)	WML-STM-001 C3; BOM lines 2, 3, 11
Manufacturing and assembly	Sawing, drilling, filing and folding thin sheet; no welding, machining or wiring; eight assembly steps with pictures	High	Arm length and tray window depend on the bought phone (confirm items 1 and 2)	WML-BLD-001; WML-DEC-001 confirm items 1 to 4
Power and controls	Tap-to-scan with the screen asleep uses 9 % of a 4,000 mAh battery a shift, with voice and the Bluetooth button; the power bank covers the screen-on case	High	The screen must sleep between scans; in heat the phone pauses charging, which tap-to-scan does not need	WML-STM-001 S4, S5, S8; WML-CAL-001 sections 5 and 13
Software and model	MobileNetV3-Large, 4.4 MB in 8-bit form, 64 to 201 ms on a low-cost phone, about 8 MB of memory; per-site grade file	High	MobileNetV3-Small stays the fallback for very old phones	WML-STM-001 M1, M2, E3; WML-CAL-001 sections 2 and 3
Data and evaluation	About 10,000 field photos (72 h of labeling) and a 2,000-image evaluation set with 400 hazards (43 h)	Low	Learning curve, coverage and hazard rarity are assumptions (M4, M5 open); hazard photos must be collected on purpose by trained people; consent and co-ownership protocol not yet written	WML-STM-001 M3 to M5; WML-CAL-001 sections 7 to 9; PR-011 not met
Human factors	With fix F1 nobody reaches up to the phone, and the result reads at 43 arcmin or more for every user from the 5th percentile up, and is spoken	Medium	Screens only; bench heights, users' heights and the voice prompt in noise to be checked with users	WML-STM-001 B1, B2; WML-CAL-001 section 12
Safety	Hazards flagged, never sorted; hazard box with sand, sharps container and taped cell terminals; rig in shade	Medium	A missed hazard is the worst failure (PR-003 at risk); larger battery packs exceed what the sand bed is sized for	WML-STM-001 C1, C2; WML-RSK-001
Supply of parts	Every station item is sold locally in Pune or online; only the reference phone is above USD 100	High	Prices are indicative except lines 12 (cited) and the option B items; the stamped scale's verification fee is to assess	BOM; WML-MKT-001

Aspect	Assessment	Rating	Risks and notes	Evidence
Partner and people	First candidate SWaCH, Pune (decided 2026-10-02), not yet approached	Medium	Without a partner there is no field data, no grade list and no evaluation	WML-PRB-001 Table 2; WML-PLN-001

Ratings: High, Medium, Low.